

Designing Experiences for the **Seattle Waterfront**



01 —

Project Vision

About the Waterfront

The Seattle Waterfront is a **gathering** place for people of all kinds—families, friends, visitors—drawn together by views and a public space.

The park promenade was established to encourage a **people-centered** space flourishing with

*“vibrant public art, inviting gathering areas, recreational opportunities, and dynamic cultural programming, creating a destination that truly **puts the community first**”*



Site Observations



The waterfront offers a highly scenic experience, yet there is an opportunity to introduce more **participatory and interactive elements** that actively engage visitors beyond passive observation.

The current sensory experience is predominantly visual, and incorporating **auditory** elements such as soundscapes or acoustic installations could create a richer, more immersive environment.



Insights

Priority 1

Preserve the scenic view

Solution

Incorporate more auditory elements

Priority 2

Foster sense of community

Solution

Introduce participatory interaction

How might we disrupt the linear promenade flow to create points of pause and interaction between visitors?

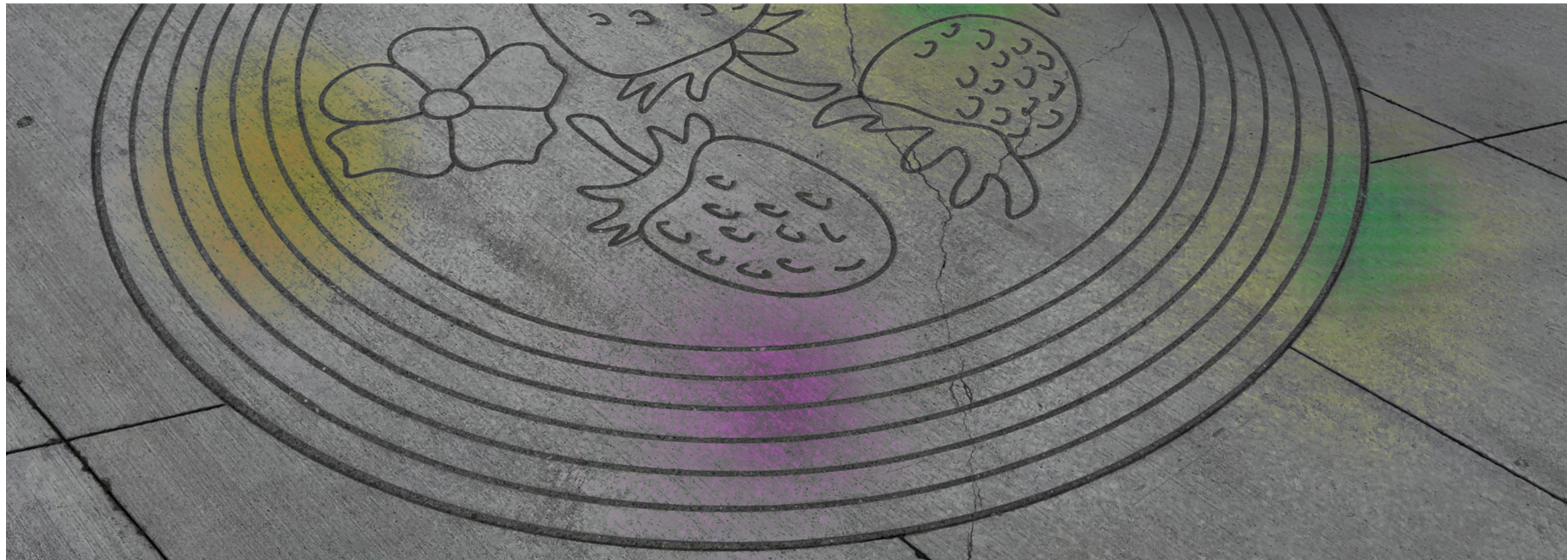
How might we enrich the waterfront's multi-sensory engagement that allow visitors to immerse themselves in the moment?

02 —

Concepts

Community Art Walkway

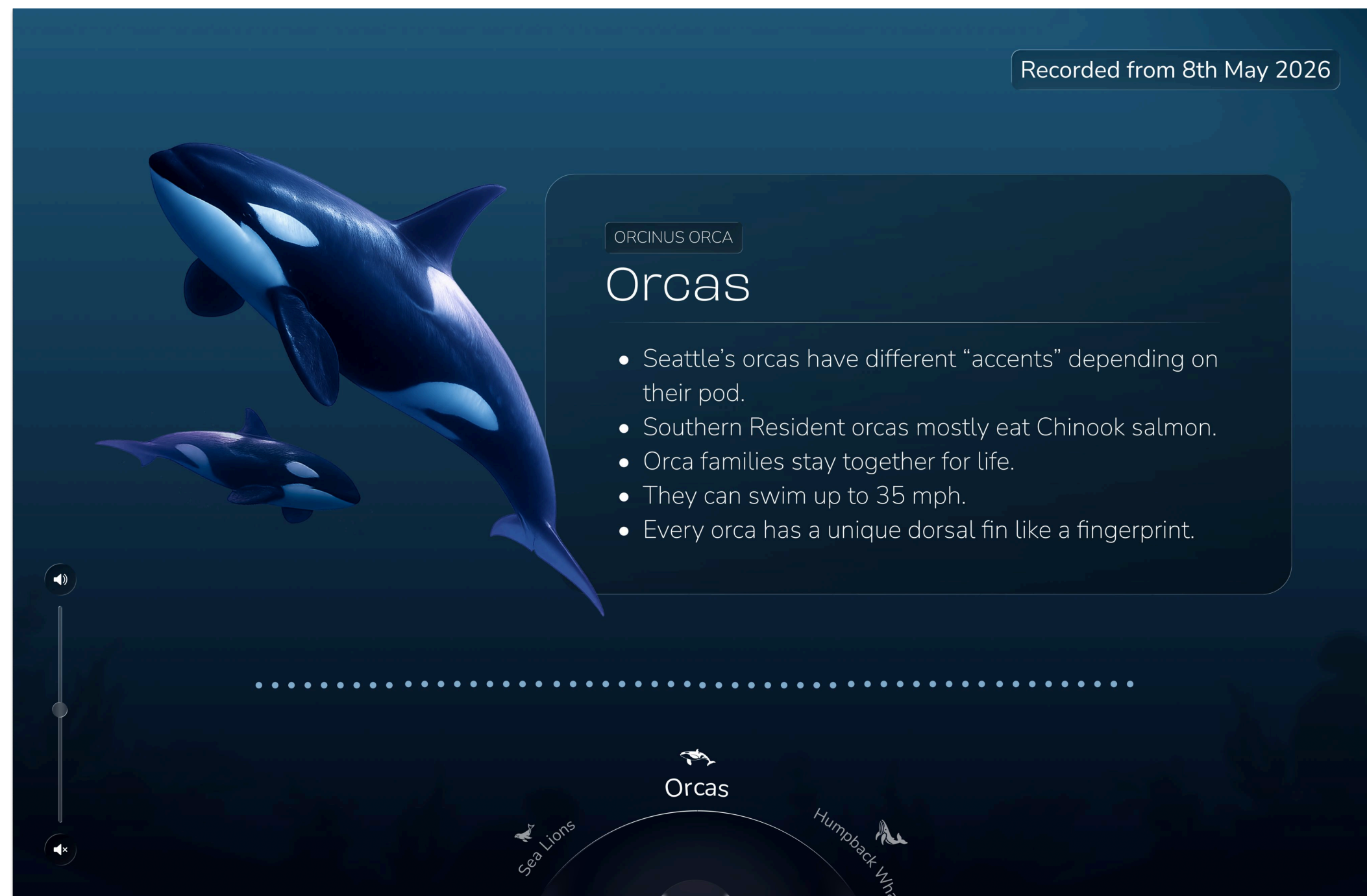
A **pressure-responsive** walkway surface embedded into the Overlook Walk that transforms pedestrian movement into a **living, collective artwork**. As visitors walk, pause, and gather, the ground beneath them becomes a canvas of light—revealing the rhythm and presence of the community in real-time.





Bioacoustic Animal Radio

An **interactive sound** installation along the Seattle Waterfront that reveals the rich but often unnoticed marine life of Puget Sound through user-guided exploration of **animal vocalizations**.



Recorded from 5th May 2026



ZALOPHUS CALIFORNIANUS

Sea Lions

- Sea lions at Seattle's waterfront are usually California sea lions.
- They can sleep floating in the water.
- Sea lions use loud barks to communicate.
- Male sea lions can weigh over 800 pounds.
- They can dive for more than 10 minutes underwater.




Sea Lions


Puffins


Orcas

How do the concepts relate?

Community Art Walkway

- connecting to the *city*, the pedestrians and artists
- primarily engages the *visual* senses
- moments of *passive creation* in society

Bioacoustic Animal Radio

- connecting to the *ocean*, the marine life community
- primarily engages the *auditory* senses
- moments of *intentional absorption* in nature

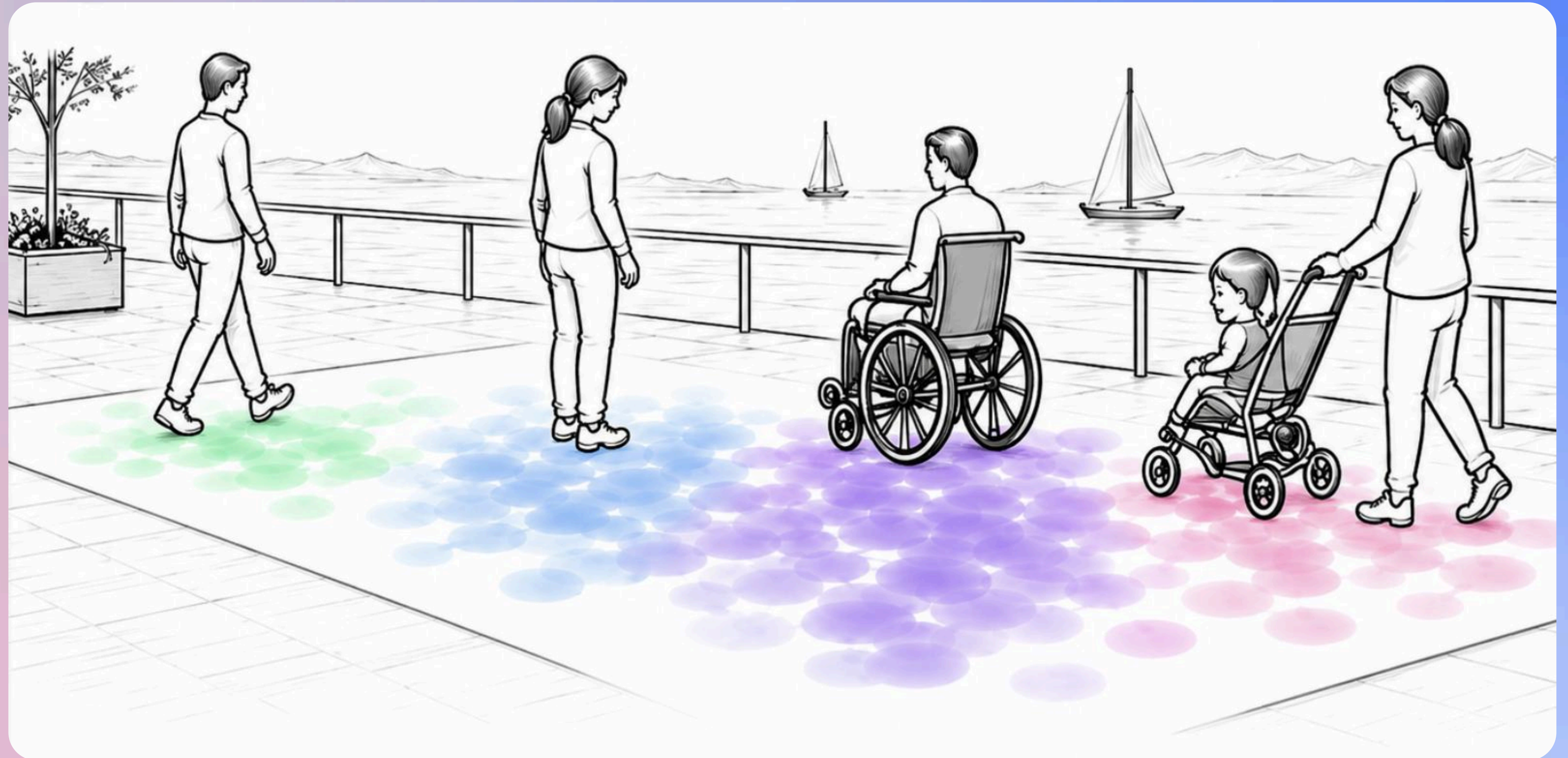
Immersion in the **Communal Essence of Seattle**
Leaning into “community” by encouraging moments of pause

03 —

Development Process

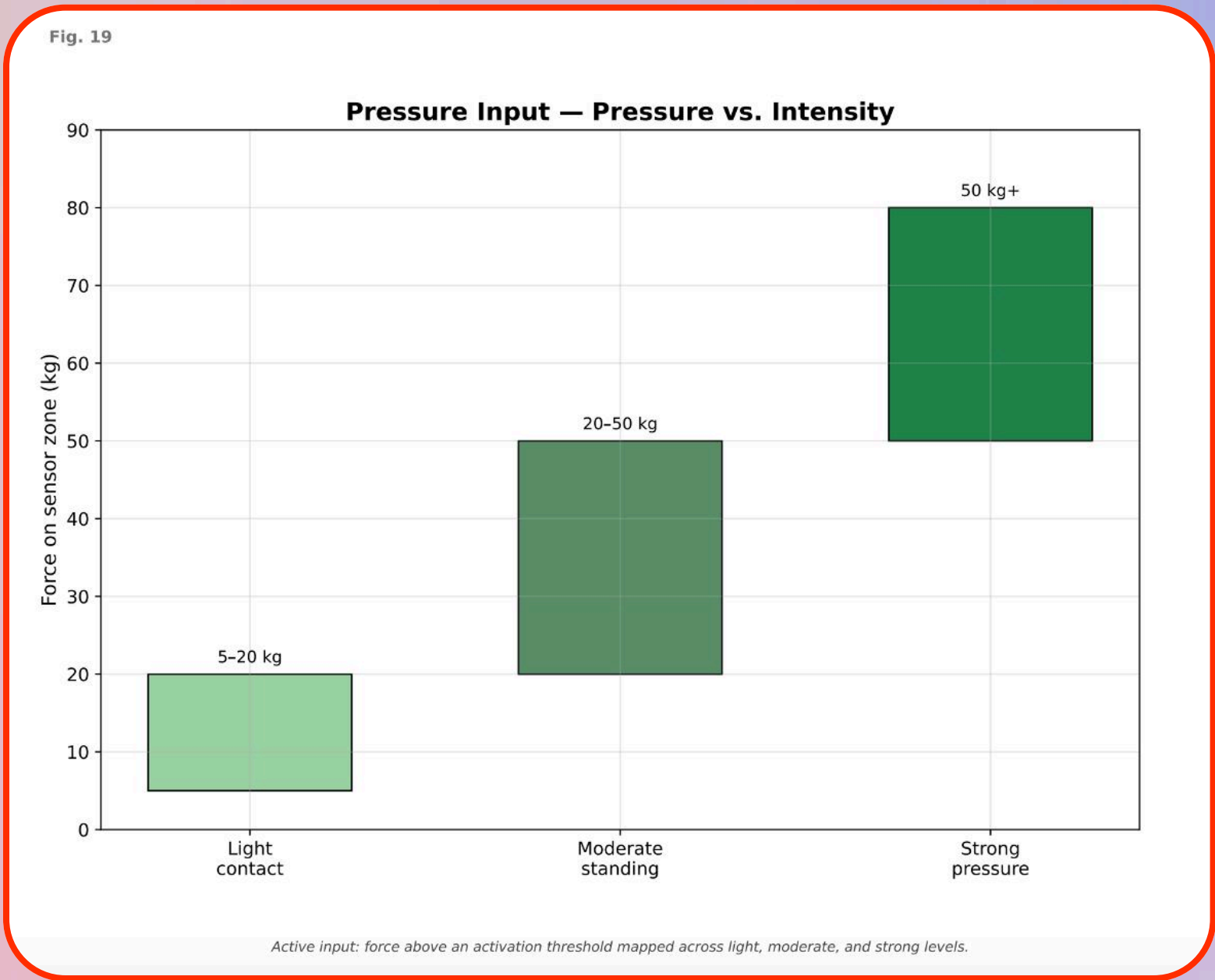
Core Interactions

- Visual Recognition
- Spatial Entry
- Tactile Exploration
- Visual Output
- Spatial Exit



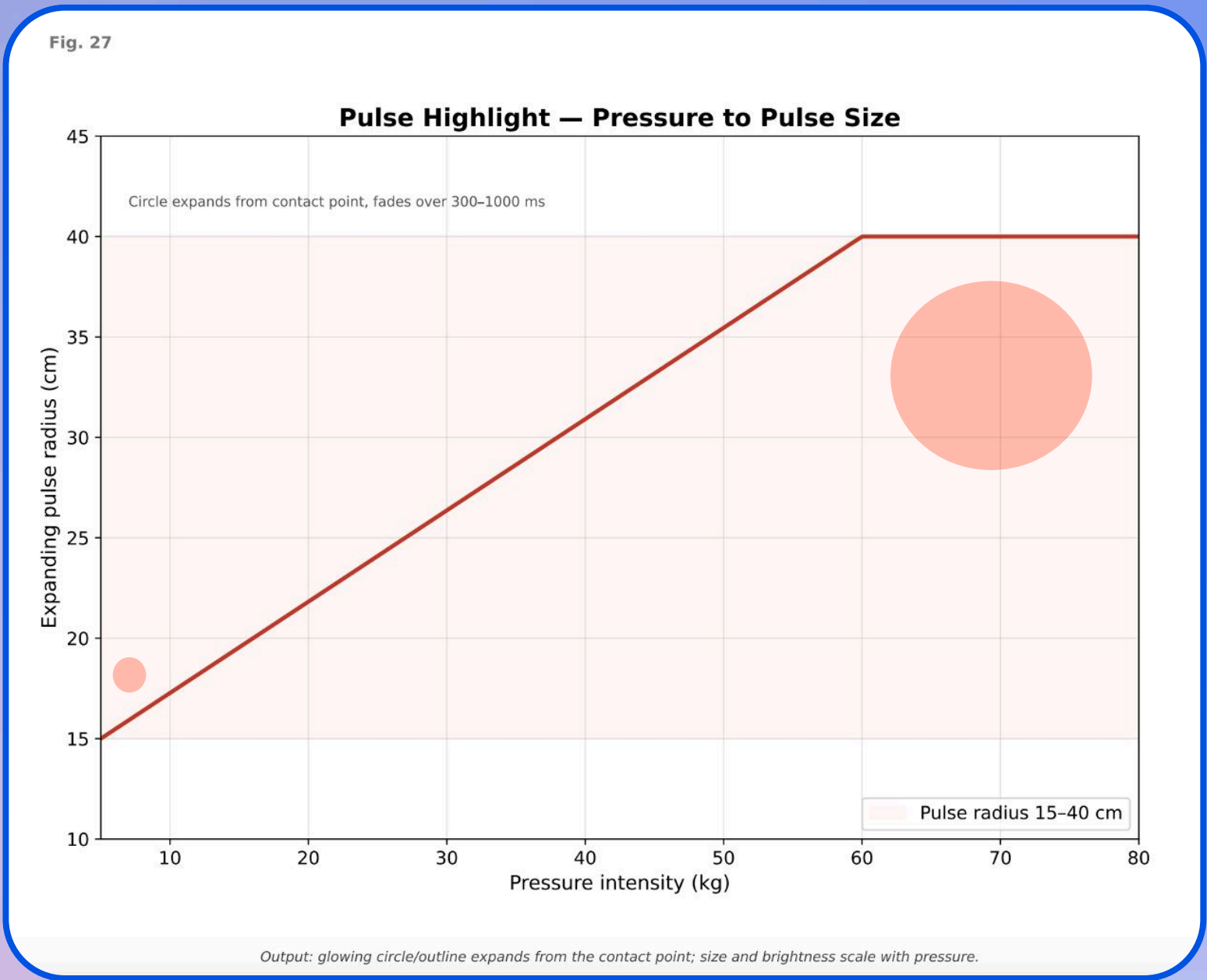
#1. Contact Confirmation Pulse

Input



Pressure

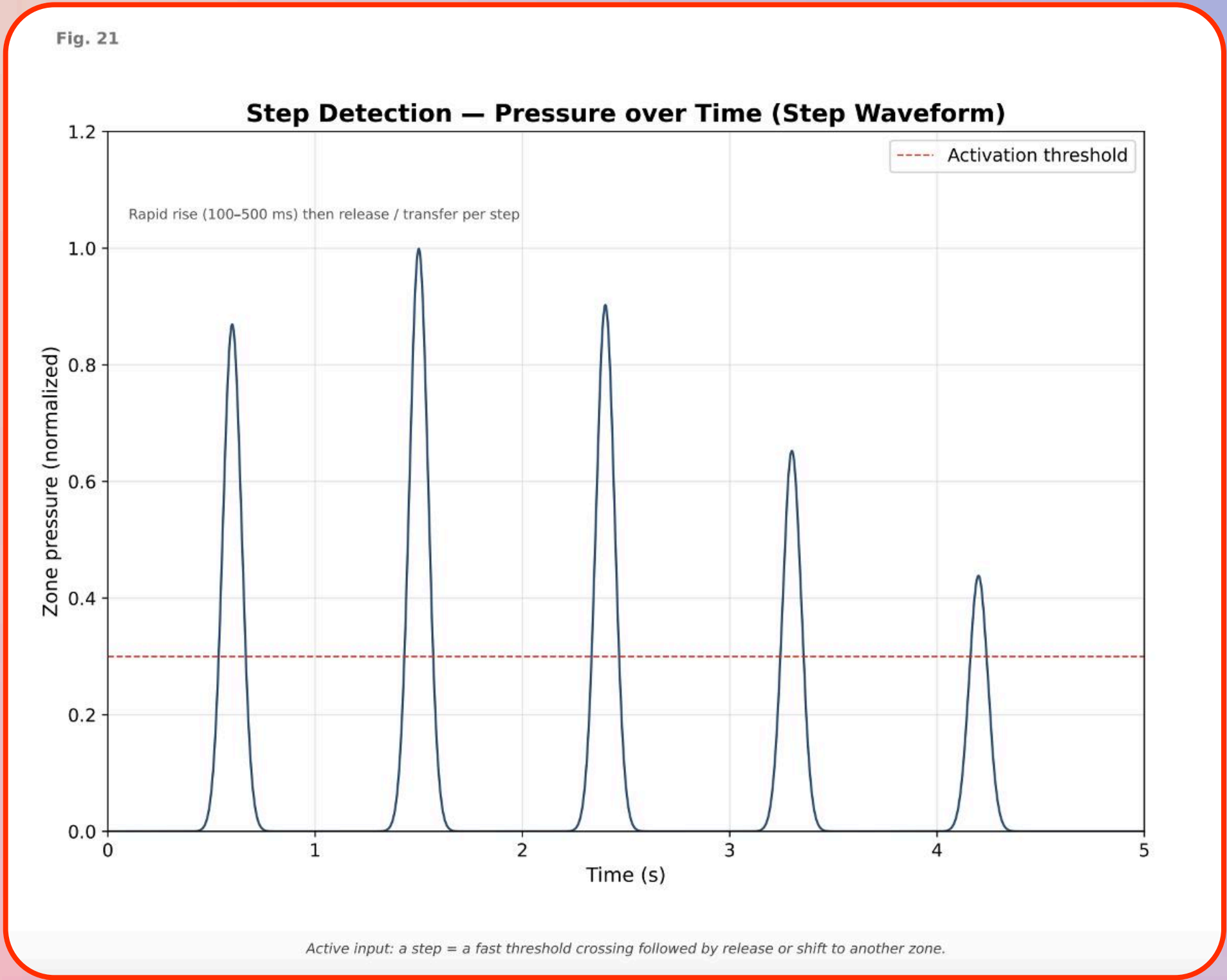
Output



Pulse Highlight

#2. Movement Trail Generator

Input



Step Detection

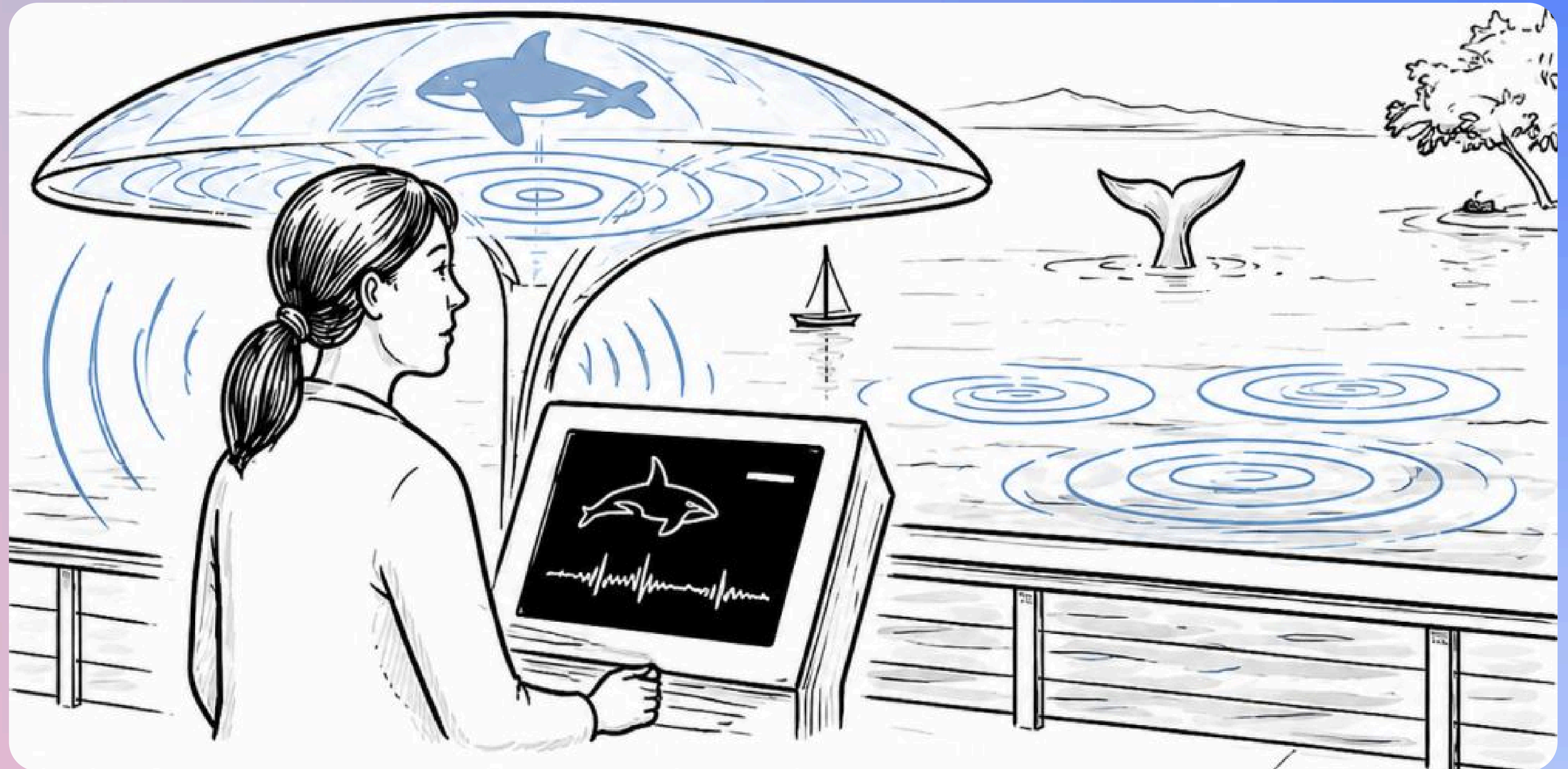
Output



Pulsing Color Trail

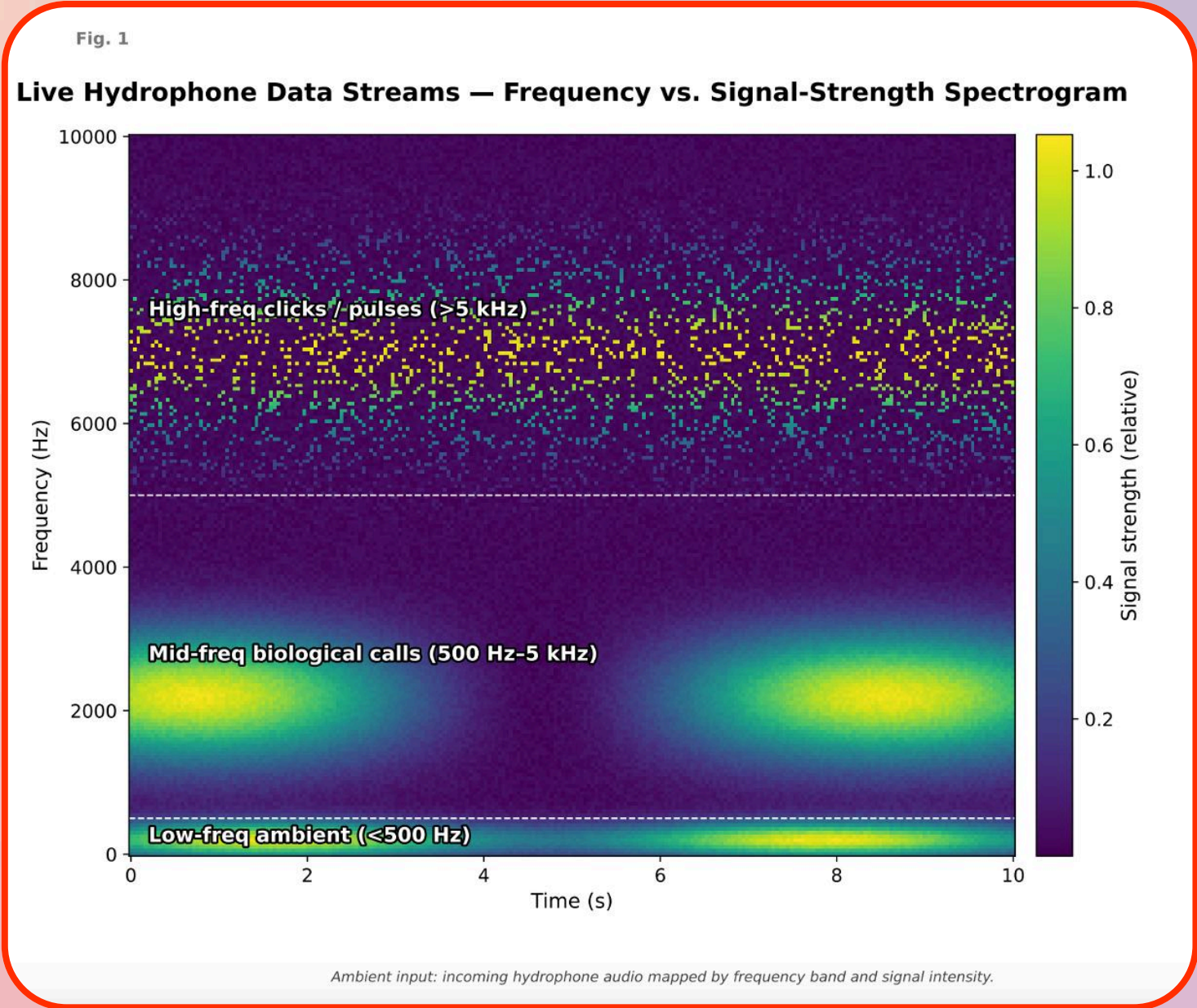
Core Interactions

- Spatial Entry
- Tactile Exploration
- Visual Output
- Immersion
- Spatial Exit



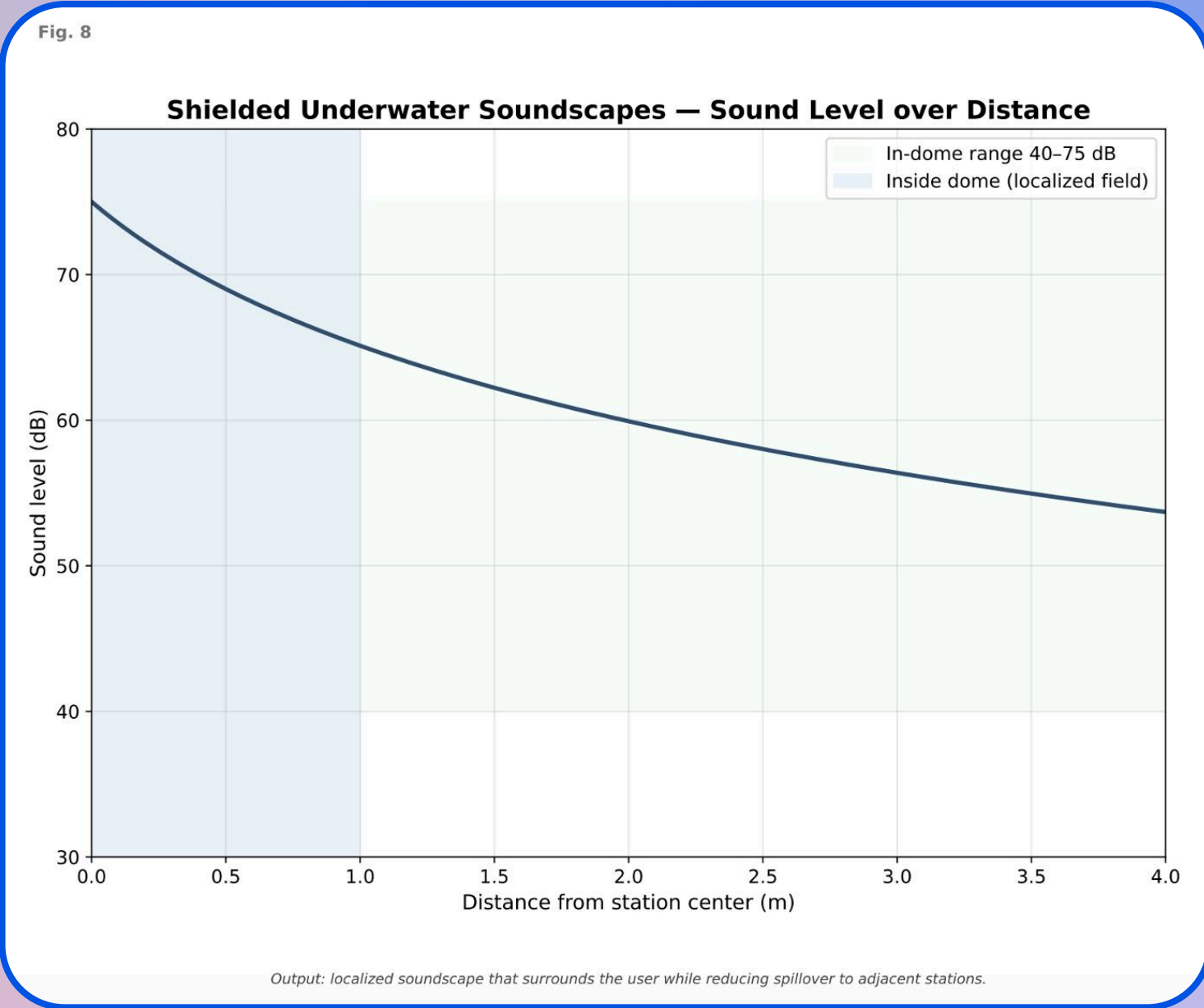
#1. Live Bio-Acoustic Interpretation

Input

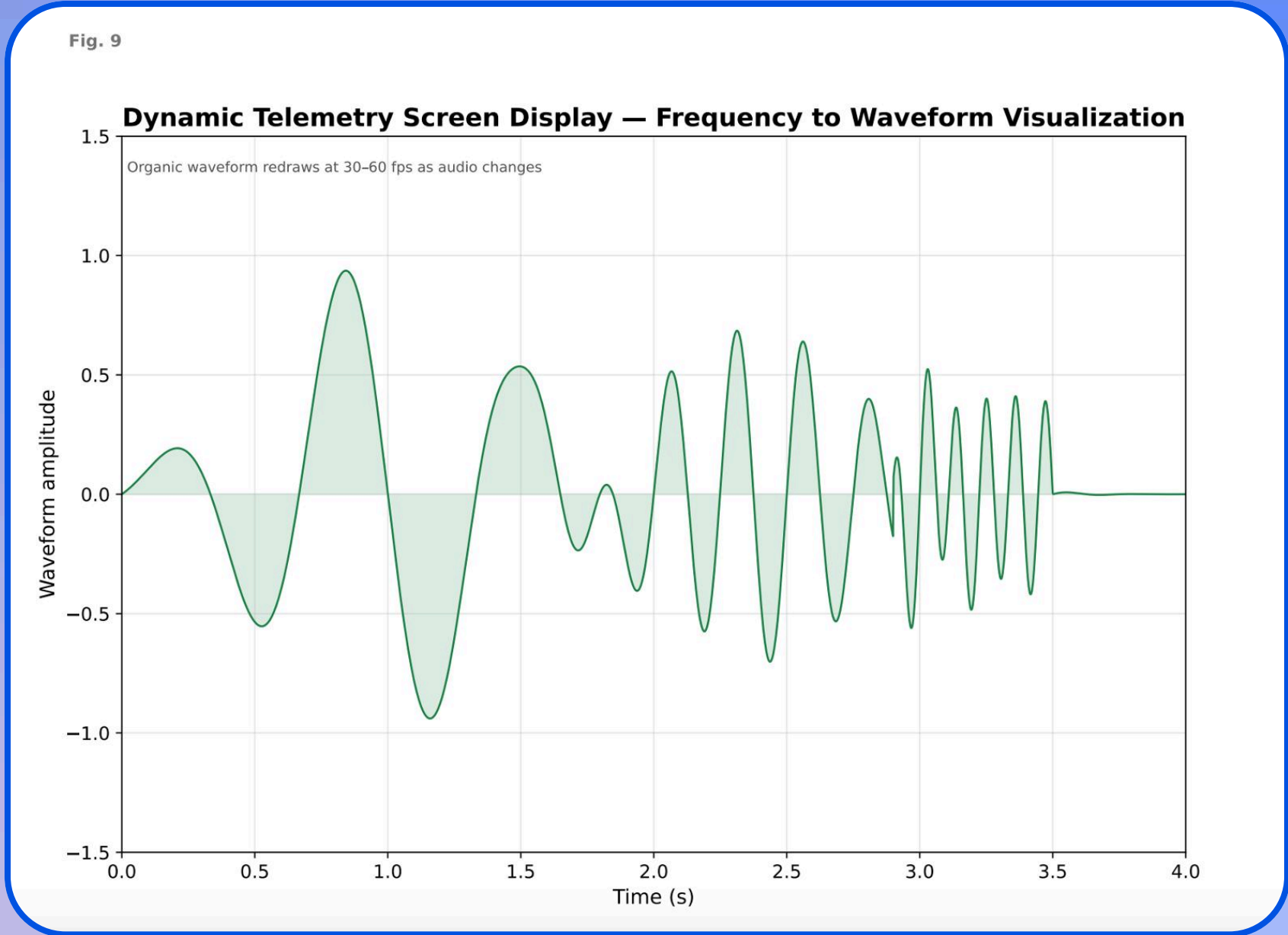


Live Hydrophone Data Stream

Outputs



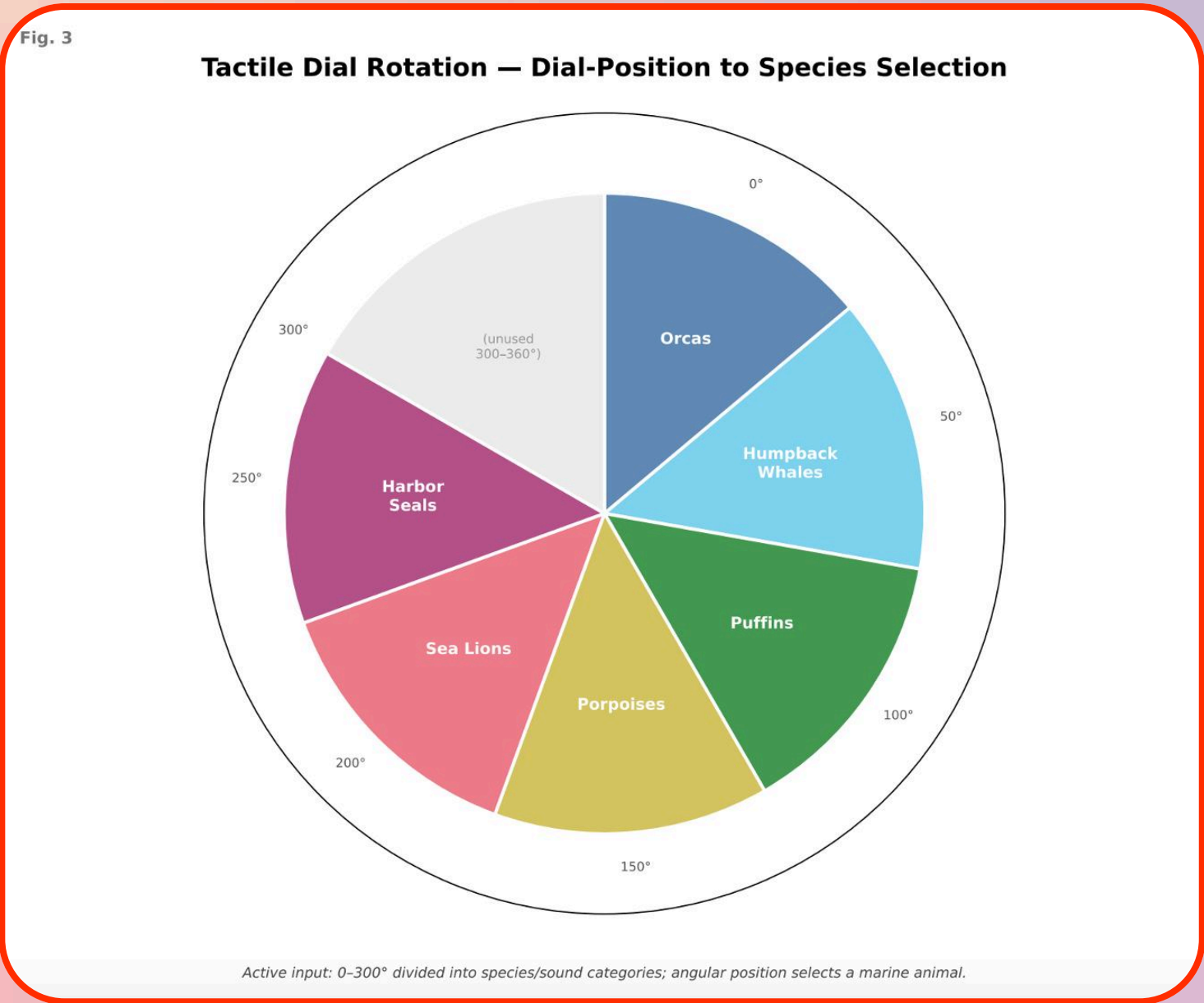
Shielded Underwater Soundscapes



Dynamic Telemetry Screen Display

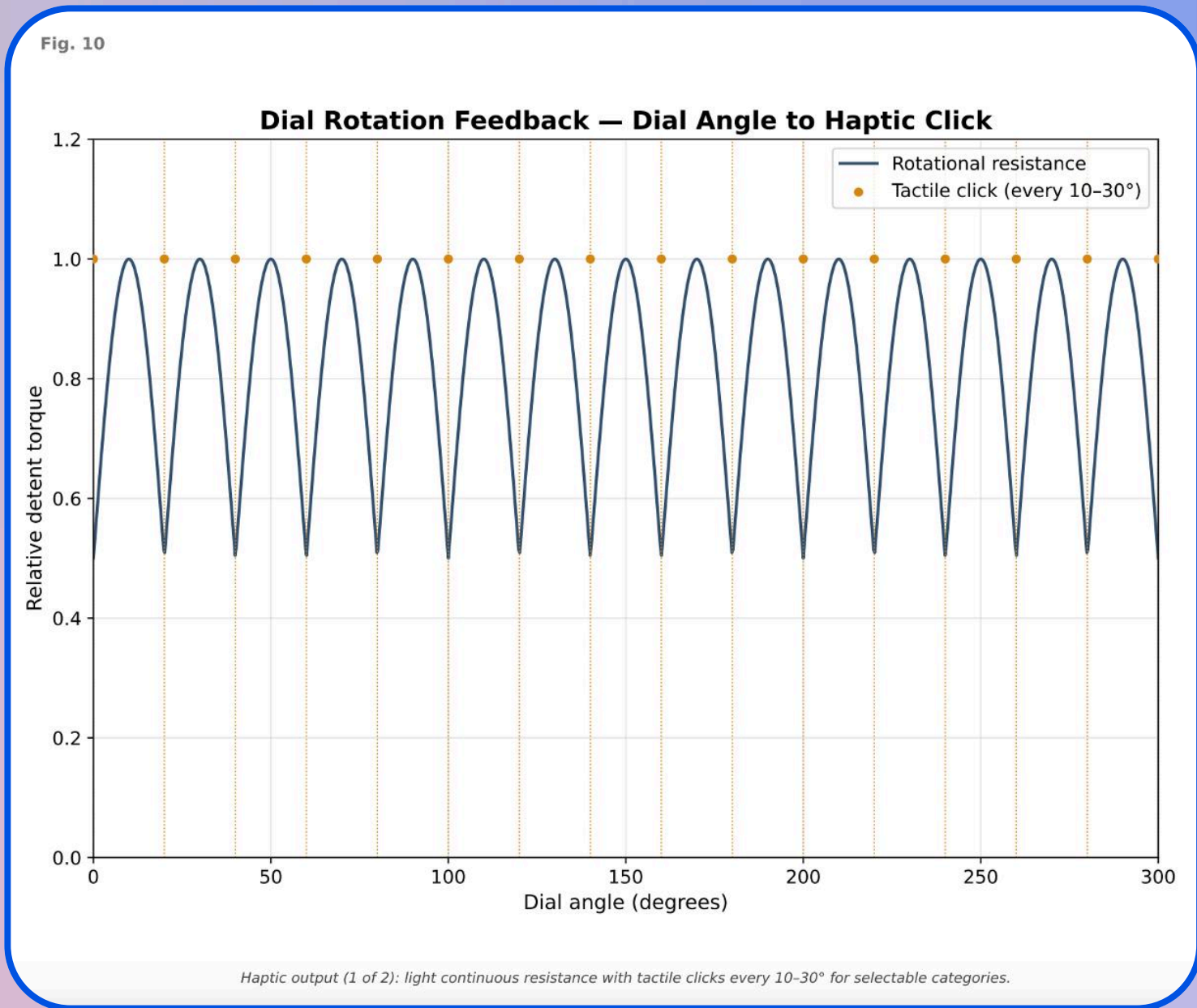
#2. Ecosystem Tuning Control

Input

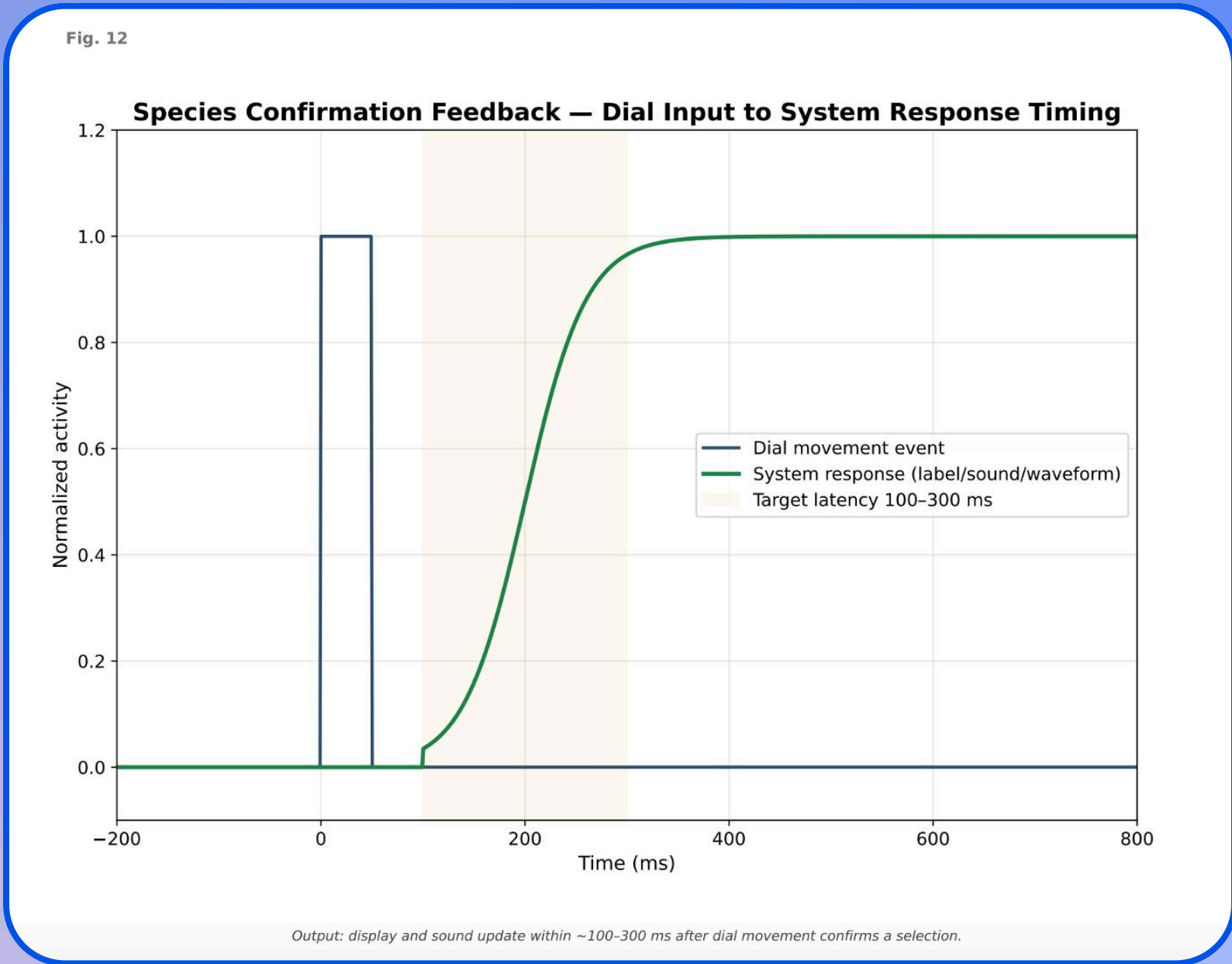


Tactile Dial Rotation:
Species View / Selection

Outputs



Dial Rotation Feedback &
Push Feedback



Dynamic Telemetry Screen
Display

Visual Direction



Moodboard

Walkway Animation

Community Art Walkway

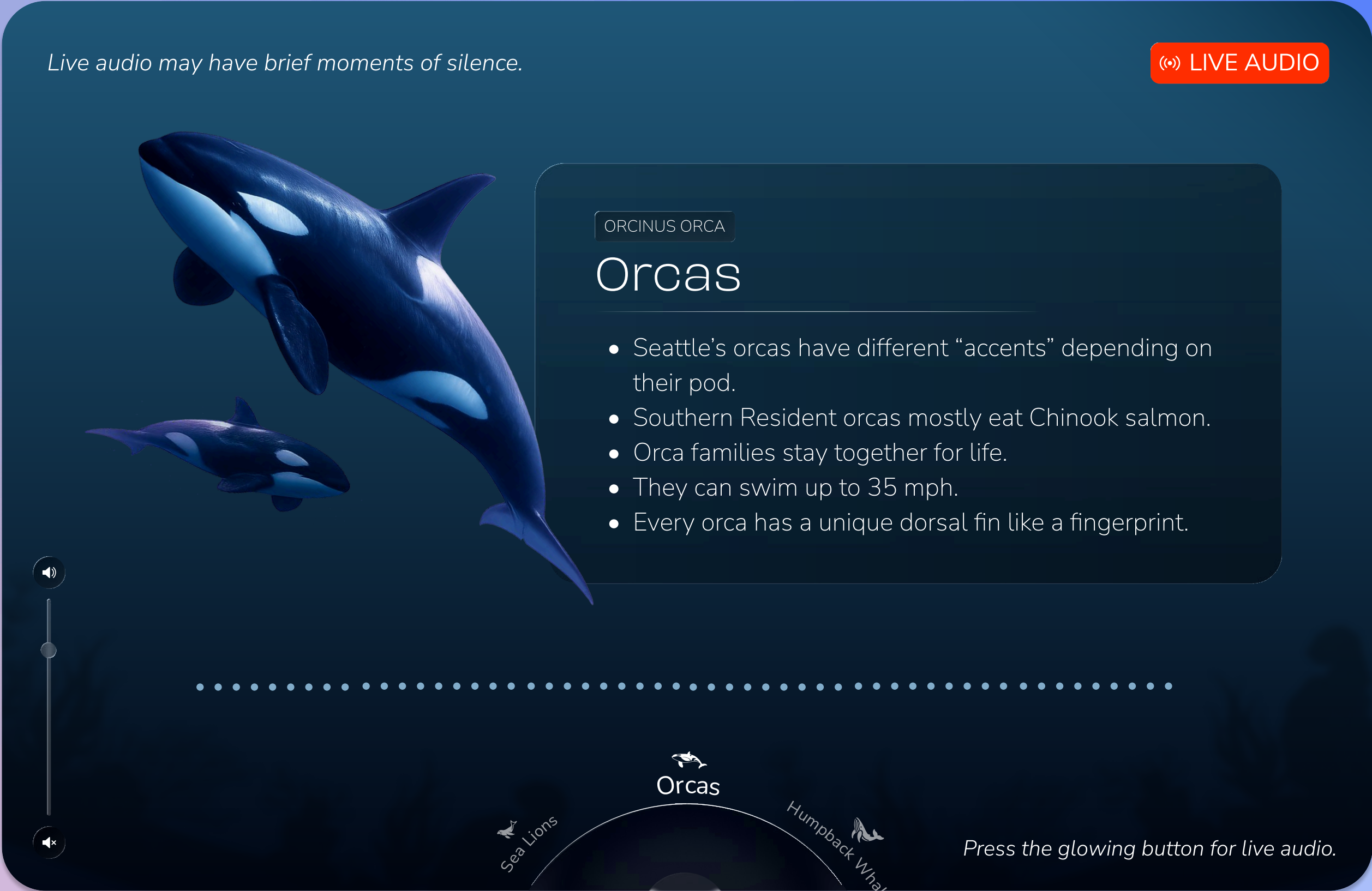
Seattle Art Inspirations



Visual Direction

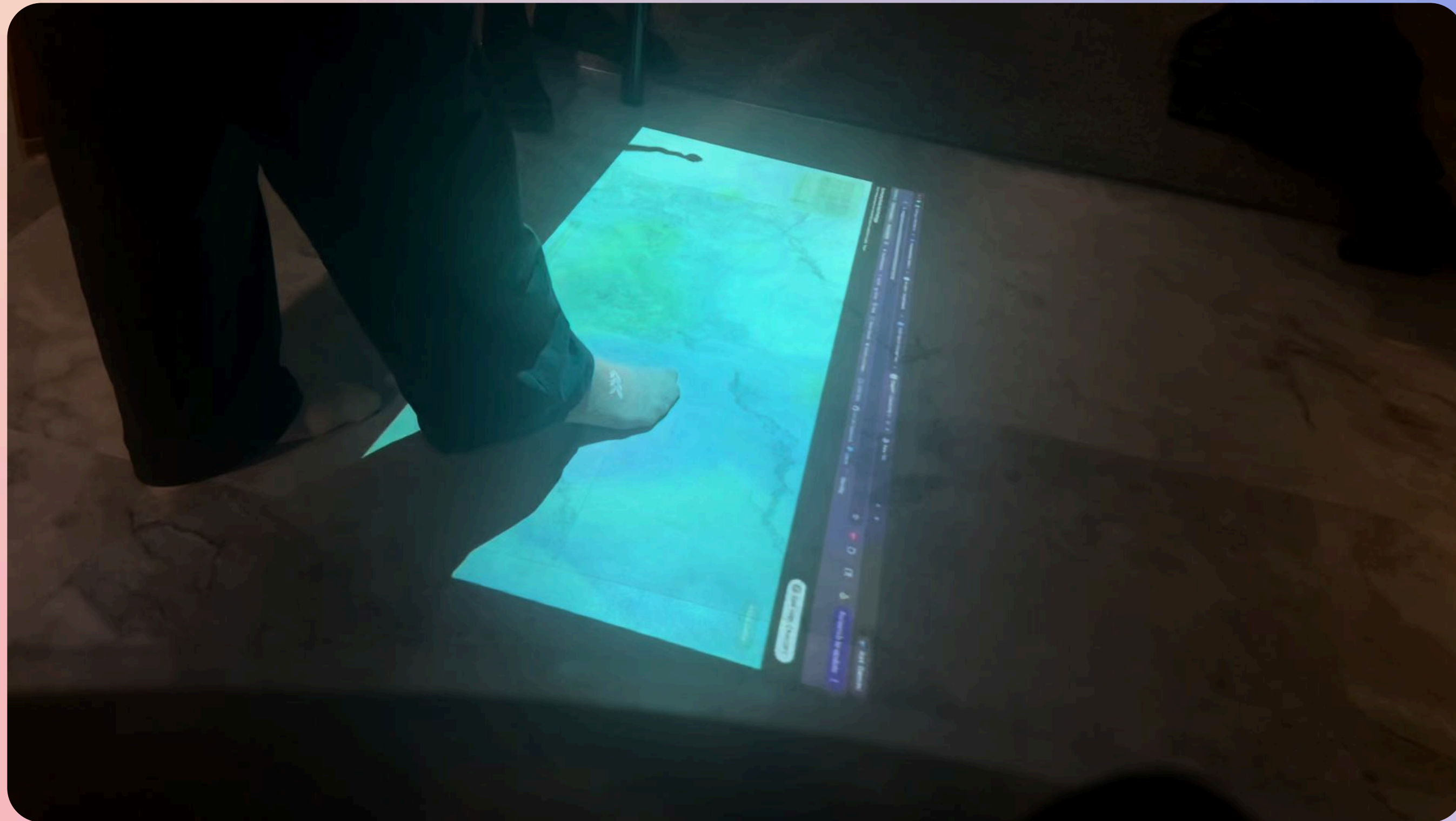


Moodboard



High-Fidelity Screen

Testing



- Built a lo-fi testing prototype to **showcase the interactions live**.
- Used **p5.js to create the visuals**, which included both colorful shapes and rippling water.
- **Projected the visuals onto the floor** for people to step onto.
- A **teammate manually simulated the trigger** using either a clicker or a laptop.
- The trigger **fired whenever a person stepped onto the projection**, demonstrating the response live.

Insights from Testing

- Users **immediately began playing with the installation** without prompting.
- **Accessibility needs consideration**, particularly for people with epilepsy.
- **Visuals should fade in slowly** rather than appearing abruptly.
- Finding inspiration from aesthetics that **already exist** in the environment (i.e., Seattle art)



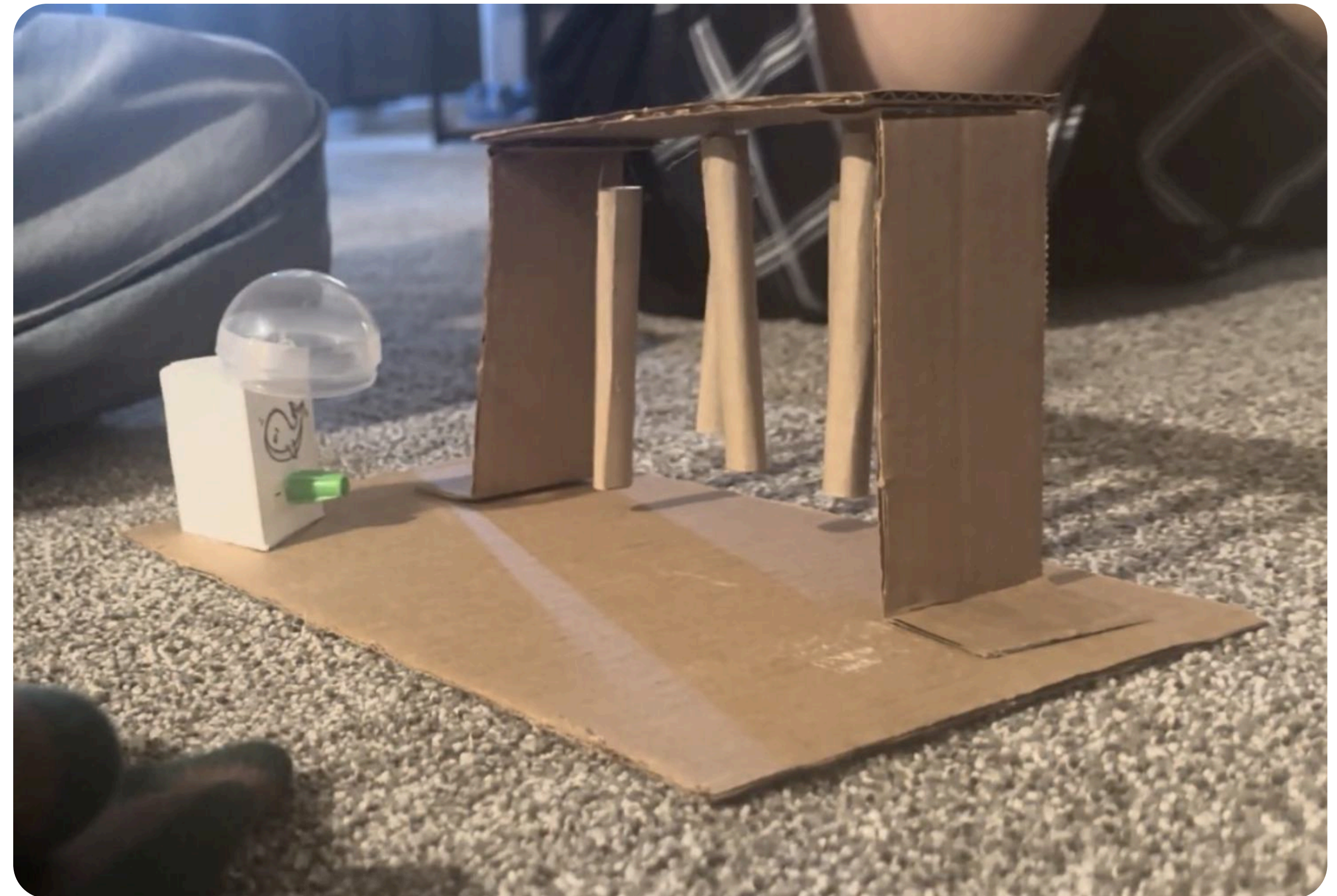
Testing



- Built a lo-fi testing prototype to **showcase the interactions in practice.**
- Created a **handmade cardboard structure to represent the chime installation.**
- Made a **folded paper structure to represent the radio** component.
- Added a **thumb tack to the paper structure to represent the dial.**
- Used a **Lego figure puppeteered by a chopstick to demonstrate how people would interact** with it.

Insights from Testing

- The **concept was well-received** and seen as cool overall.
- The **dome should ideally fit more than one person** at a time.
- Engagement ideas worth exploring **include a hologram element**.
- The **dome caused initial confusion**, felt disconnected from the chimes.
- The **chimes sit too low to be a safety hazard**.



03 —

Video

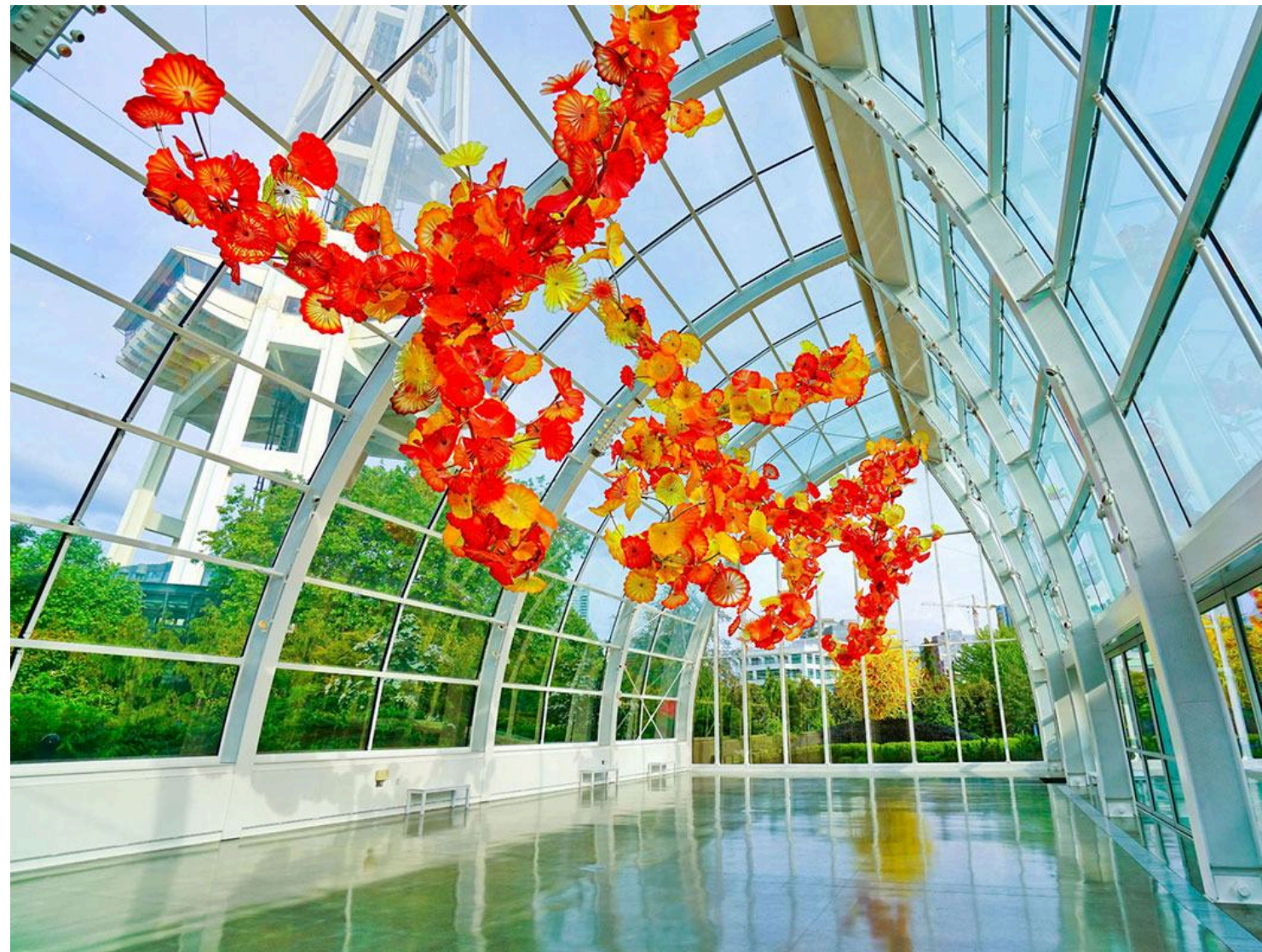
A photograph of the Seattle waterfront with a city skyline in the background. In the foreground, a person is skateboarding on a paved area, and a red trash can is visible on the right. The sky is blue with some clouds.

SEATTLE WATERFRONT THE COMMUNAL ESSENCE OF SEATTLE

Critical Analysis

City of Seattle's Values & Civic art

This concept brings light to the work of **local artists**, joining strangers together to form **collaborative art**. It benefits mobile, non-visually impaired waterfront visitors and local artists.



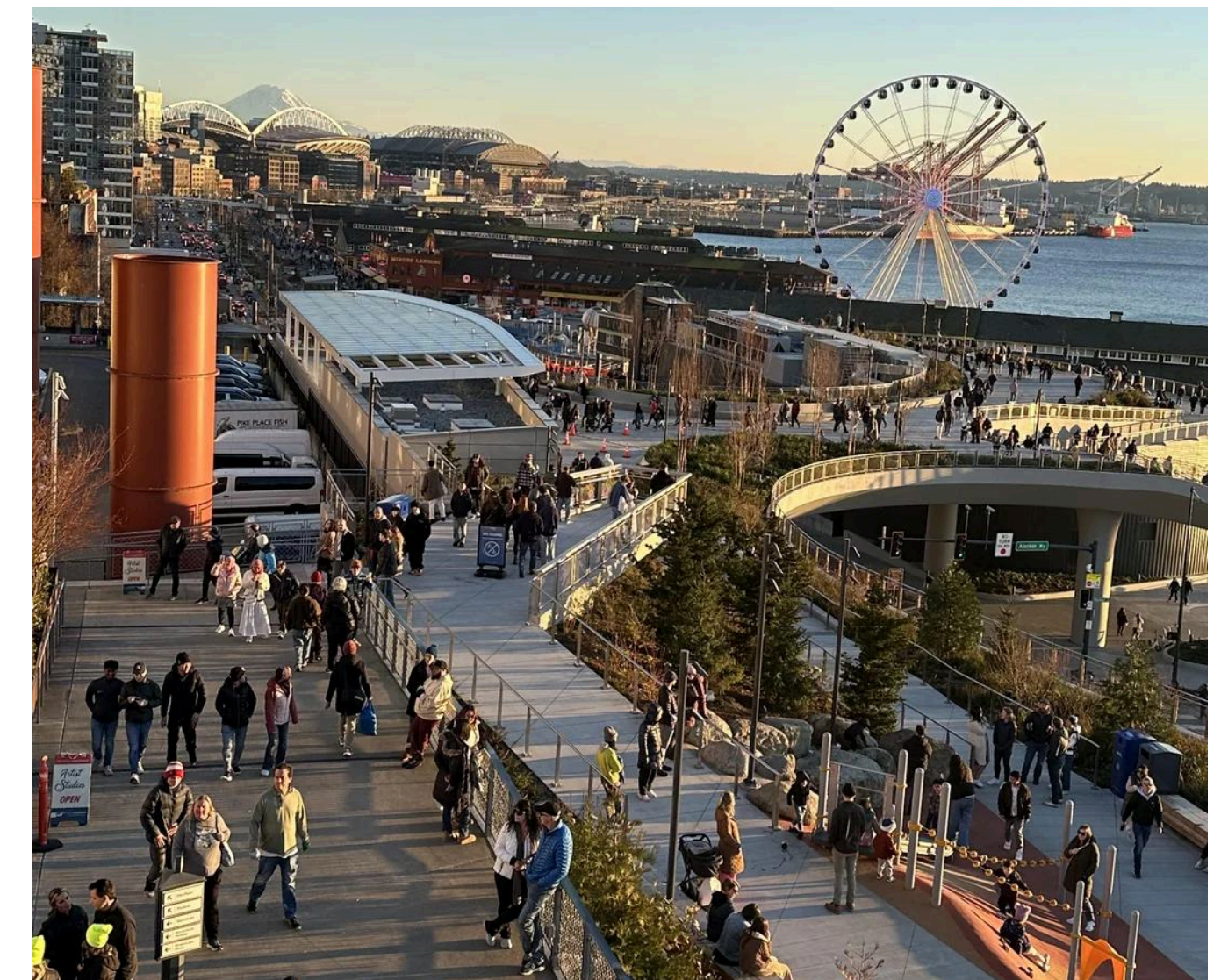
Possible Excluded Populations

- People with visual impairments or low mobility
- Artists whose aesthetics do not fit the medium of an interactive walkway
- People who feel afraid or embarrassed about being seen interacting with it.



Cases of Misuse

- Crowd congestion
- The weather negatively affects the floor (ex: rain causes slippery floor)
- Floor cracks under pressure



Critical Analysis

Environmental & Ecological Values

This concept optimizes for **emotional connection to wildlife**. It benefits marine ecosystem advocacy, education groups, and curious waterfront visitors.



Possible Excluded Populations

- People with hearing impairments
- People with visual impairments
- People with physical limitations (ex: wheelchair users)



Cases of Misuse

- People refuse to leave the pod and hog the experience
- It detracts from being present in the moment
- The system produces misidentifies the animals



Thank you!

